An Automatic Method for Epileptic Seizure Detection Based on Deep Metric Learning

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***Abstract*—Electroencephalography (EEG) is a commonly used clinical approach for the diagnosis of epilepsy which is a life-threatening neurological disorder. Many algorithms have been proposed for the automatic detection of epileptic seizures using traditional machine learning and deep learning. Although deep learning methods have achieved great success in many fields, their performance in EEG analysis and classification is still limited mainly due to the relatively small sizes of available datasets. In this paper, we propose an automatic method for the detection of epileptic seizures based on deep metric learning which is a novel strategy tackling the few-shot problem by mitigating the demand for massive data. First, two one-dimensional convolutional embedding modules are proposed as a deep feature extractor, for single-channel and multichannel EEG signals respectively. Then, a deep metric learning model is detailed along with a stage-wise training strategy. Experiments are conducted on the publicly-available Bonn University dataset which is a benchmark dataset, and the CHB-MIT dataset which is larger and more realistic. Impressive averaged accuracy of 98.60% and specificity of 100% are achieved on the most difficult classification of interictal (subset D) vs ictal (subset E) of the Bonn dataset. On the CHB-MIT dataset, an averaged accuracy of 86.68% and specificity of 93.71% are reached. With the proposed method, automatic and accurate detection of seizures can be performed in real time, and the heavy burden of neurologists can be effectively reduced.**

***Index Terms*—Deep learning, electroencephalography (EEG), epilepsy, metric learning.**

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I. Introduction

NEARLY 50 million people suffer from epilepsy worldwide according to the World Health Organization (WHO) [1]. Although most patients become seizure-free with antiepileptic drugs, there are still 20–40% of patients that do not respond to antiepileptic drugs [2]. Therefore, surgery may be an option for these patients with intractable epilepsy and accurate detection of seizures is necessary for the pre-surgical evaluation. Besides, real-time detection can also provide timely alarms about ongoing seizures. Electroencephalography (EEG), as a record of brain electrical activity, is widely and commonly used in the diagnosis of epilepsy. For some patients with intractable epilepsy, abnormalities do not register on scalp surface electrodes thus intracranial EEG (iEEG) has to be used [3]. Now, the analysis of EEG recordings is still mainly carried out by experienced neurologists which can be time-consuming and not possible in real-time applications. To make the detection of seizures faster and more accurate, and to reduce the burden of neurologists, automated seizure detection from EEG signals has long been an active research topic [4]–[7]. Due to the low signal-to-noise ratio (SNR) [8] and the non-stationary nature of EEG signal [9], seizure detection is a challenging problem as well as other EEG-based automatic diagnostic and detection.

In machine learning, seizure detection is considered as a classification problem and the corresponding algorithms follow a standard paradigm: preprocessing including artifact removal, filtering, and segmentation, then feature extraction, and finally classification. The work in [10] segmented the iEEG signals and used Hilbert-Huang transform (HHT) for feature extraction and Bayesian classifier for classification. Similarly in another study [11], EEG signals were first decomposed using wavelet packet decomposition and re-constructed into commonly used intrinsic frequency bands. Then the fractal spectrum feature describing the dynamical fractal structure was extracted from several sub-bands of EEG signals and then classified by support vector machine (SVM). The method in [12] utilized Stockwell transform for feature extraction and K-nearest neighbor (k-NN) classifier for classification. In [13], a novel machine-learning algorithm was proposed for end-to-end prediction of antidepressant response in major depression. The model consists of spatial filtering, extraction of band power feature, and linear regression, which were trained altogether via convex optimization. The algorithm performed well across datasets. Similarly in [14], a generalizable algorithm for identification of psychiatric disorder subtypes from EEG functional connectivity patterns was established.

In the past few decades, deep learning has rapidly arisen and showed significant performance in many areas including computer vision (CV) and natural language processing (NLP). Convolutional neural network (CNN) is one of the most successful categories of deep neural networks and has outperformed hand-crafted features and traditional classifiers on image classification and many other tasks [15]–[17].

In recent years, there was an emerging trend of applying deep learning methods to the analysis of EEG signals, including epilepsy diagnosis and seizure detection [18]. The method in [19] got rid of the standalone feature extraction step which is crucial in traditional machine learning pipelines, and directly fed raw EEG signals into bidirectional Long Short-Term Memory (LSTM) for detection of seizures. The work in [20] also skipped feature extraction and classification was done by a Gated Recurrent Unit. Seizure detection on raw EEG signals was done in [3] using a CNN. The method in [21] performed attention-based seizure detection and patient detection based on raw EEG signals via CNNs as well. Using autoencoder, the EEG signal was separated into patient-specific and patient-independent parts in the latent space. Likewise, [22] utilized adversarial domain generalization to eliminate the subject variability of deep feature for accurate emotion recognition. In contrast, 2-dimensional features were obtained via short-time Fourier transform (STFT) and classified using CNN in [23], leveraging both traditional feature extraction and deep learning. The method in [24] used the common spatial pattern (CSP) as a low-dimensional feature for classification through shallow CNN.

As stated in many of the above-mentioned works, it is the relatively small size of EEG datasets that severely limited both the performance and broader application of deep learning methods which are data-demanding. A straightforward workaround is data augmentation. The algorithm in [24] created artificial EEG signals by concatenating training segments. Another study [25] used a similar but improved approach in which the concatenation was performed in the frequency domain as well as the temporal domain. A generative adversarial network (GAN) was employed in [26] to generate artificial signals for training. Concatenation of samples is too simple and generative models are data-demanding themselves and therefore not very helpful as well.

Besides data augmentation, there exist a large number of other strategies in deep learning addressing the lack of data or so-called few-shot problem, including deep metric learning [27], [28], meta-learning [29]–[31] and multitask learning [32]. The algorithm in [33] borrowed the idea of triplet loss which is broadly employed in face identification into EEG classification for brain-computer interface. A similar model was adopted in [34]. The method in [35] used transfer learning and contrastive loss for patient-independent seizure prediction, and achieved improved performance on the CHB-MIT dataset. Similarly, [36] utilized a modified version of Siamese networks and the similarity scores between the given sample and support samples were used at inference time, in a meta-learning setup. Some enlightenment was given but the potential of the methods is still far from being fully exploited.

In this paper, we proposed a deep metric learning model for automatic seizure detection inspired by the Siamese Networks [27]. The performance of the method is first evaluated on ictal and interictal subsets of the Bonn University Dataset [37], which is a benchmark dataset that various methods had been tested on. Then, the method is further verified on a larger dataset, CHB-MIT, which is continuously recorded and multichannel, for more realistic evaluation. Because EEG recordings are one-dimensional time series, single-channel and multichannel one-dimensional convolutional embedding modules are designed accordingly for the two datasets. In addition, deep metric models learn to map inputs into embedding spaces but does not directly output the classification result. Therefore, a stage-wise training strategy is used incorporating additional classification layers. Afterwards, a majority vote strategy is also employed and further improved the performance.

The rest of the paper is arranged as follows. In Section II, our proposed method is explained in detail, including preprocessing, the designed deep metric model, and the training strategy. Section III presents experimental results of our proposed method, including comparison with baseline methods, and comparison using different metric functions and classifiers. In Section IV, discussion and comparison with recent existing methods are made. In Section V conclusion is drawn and possible future works are suggested.

II. Methodology

The proposed method consists of a deep metric learning model and its corresponding training strategy. The model mainly consists of two identical embedding modules with shared weights, along with a metric module. The embedding modules for single-channel and multichannel EEG are inevitably different. Fig. 1 shows the model's architecture with embedding module for single-channel EEG. The architecture for multichannel EEG is different only in the embedding module. A pair of samples and their labels are fed into the model at a time. Each sample is passed through one of the two identical embedding modules and the normalized embedding vector in a relatively low dimensional embedding space is generated as would be stated below. Then, the weighted distance between the two embedding vectors is calculated by the metric module to decide whether they are similar or dissimilar indicating that they are from the same class or different classes. Binary cross-entropy loss is computed between predicted similarity and ground-truth similarity and is used for the update of model parameters through backpropagation.

In the subsections below, the datasets being used and our preprocessing method is first introduced. Then the proposed metric learning model is explained in detail, including the embedding modules and the metric module. Finally, our training strategy and majority vote strategy are detailed.

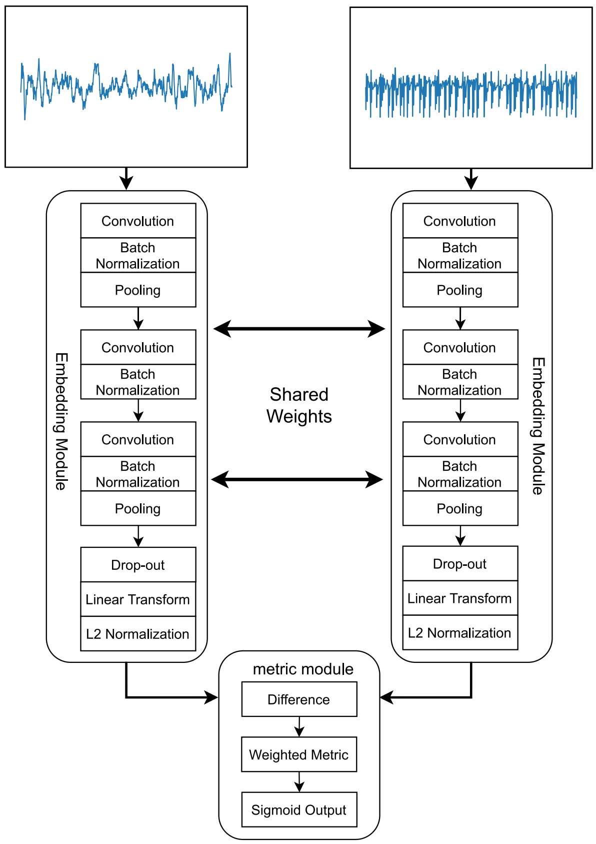

Fig. 1. Overall structure of our proposed deep metric learning model for single-channel EEG signal. The two waveforms in the figure are randomly picked from the Bonn Dataset for demonstration purpose.

A. Datasets and Preprocessing

The first dataset used in this work was made public by Bonn University [37] (http://epileptologie-bonn.de/cms/front_content.php?idcat=193) and has been used as a benchmark on epilepsy seizure detection [38]. The dataset consists of 5 subsets namely Set A to E, also known as Set Z, O, N, F, and S, each containing 100 EEG signals of length 23.6 seconds, or equivalently 4097 sample points given the sampling rate is 173.61 Hz. Set A and B are scalp EEG taken from five healthy volunteers and Set C through E are intracranial EEG from five patients. Signals from Set C and D were recorded when there was no seizure activity and are therefore considered interictal. Among them, signals in Set D were recorded from the epileptogenic zone whereas signals in Set C were recorded from the hippocampal formation of the opposite hemisphere. Signals of Set E are ictal as they were recorded during seizure activity. C versus E and D versus E classification problem is chosen for our experiments in this paper except otherwise mentioned, as they are both intracranial EEG signals. The data have the spectral bandwidth of the acquisition system, which is 0.5 Hz to 85 Hz. A 40 Hz low-pass filter suggested by the website hosting the dataset is not applied in this work, because the model is supposed to be learning as much information as possible from the available data. The EEG signals from the chosen subsets are directly used in our work without additional filtering. Any signal is segmented into 5 parts using a sliding window of length 2048 and 75% overlapping, discarding the last point. Corresponding labels are inherited from the original signal. This is done on the training set and test set separately.

The second dataset, CHB-MIT, was collected from the Children's Hospital Boston by [39] (downloadable from PhysioNet [40] https://physionet.org/content/chbmit/1.0.0/). Multichannel scalp EEG signals were recorded from 23 subjects with intractable seizures with sampling rate of 256 Hz. For each subject, there are multiple recordings that were continuously recorded for hours, with the beginning and end of seizures annotated by clinical experts. 18 EEG channels that all subjects have, including FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ and CZ-PZ are chosen. Several recordings of subject *chb12* were not originally bipolar referenced and are re-referenced offline using MNE-Python toolbox [41]. As scalp EEG has lower SNR compared to intracranial EEG, a 5th order Butterworth band-pass filter from 0.5 Hz to 70 Hz, and a notch filter of 60 Hz are applied to reduce the influence of noise. Artifacts are also screened and removed. Signals annotated as seizures are considered ictal. Signals that are more than 1 h before and after any seizure activity are considered interictal. Any signal is segmented using a sliding window of length 2048 sample points. 75% overlapping is used for ictal signals to create more training segments, as the data of this class is significantly less then interictal data. Afterwards, interictal signals of each subject are randomly discarded to maintain the ratio to ictal signals to be 1:1.

B. Proposed Embedding Module

1) Single-Channel Embedding Module: Our proposed embedding module for single-channel EEG signal is a sequential neural network that mainly consists of one-dimensional convolutional layers, max-pooling layers, and a linear layer, as shown in any of the two embedding modules in Fig. 1. Convolutional layers are excellent at deep feature extraction of spatial or temporal data, and one-dimensional convolutional layers are widely adopted for EEG signal processing, empirically outperforming recurrent neural networks like LSTM. Rectified linear unit (ReLU) is used as activation of the convolutional layers. L2 regularization is applied to convolution kernels to regularize the distribution of trainable weights, thus helps to prevent over-fitting. A technique called batch normalization is applied after each of the convolutional layers to tackle the problem of internal covariate shift and stabilize the training. Max-pooling layers are used to down-sample the extracted features to reduce the redundancy while keeping the most discriminative parts. After the last max-pooling layer, a drop-out layer that randomly drops out each component of the feature vector at a given rate is inserted. Though drop-out layers are commonly applied after fully-connected layers, they can still be used as a way of regularization in our case where it is appended after a max-pooling layer. Deep features extracted by the layers are then transformed into an embedding space through the linear layer. L2 normalization is applied afterward,

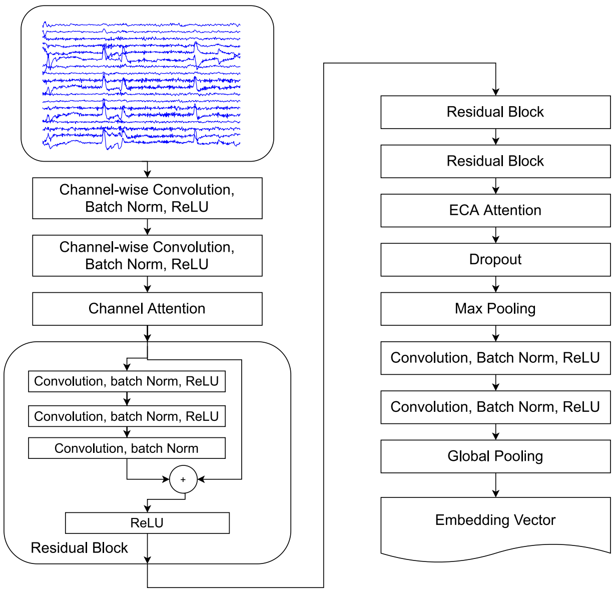

Fig. 2. Architecture of the embedding module designed for multichannel EEG signal. The waveform in the figure is randomly picked from the CHB-MIT Dataset for demonstration purpose.

projecting the embedding vectors onto the surface of a unit hyper-sphere, which is empirically found to be helpful.

2) Multichannel Embedding Module: An improved embedding module is proposed to handle multichannel scalp EEG signals which is more commonly used. Shown in Fig. 2, raw signal is first feed into two channel-wise convolutional layers one after another for the extraction of intra-channel feature. For each channel-wise convolutional layer, any input channel is mapped into 2 output channels individually, followed by batch normalization and $ReLU$ activation, which are also applied after every conventional convolutional layers afterwards. Afterwards, channel attention mechanism is implemented for automatic channel selection, using Squeeze-and-excitation block [42], following which three bottleneck residual blocks are stacked. Each residual block consists of a convolutional layer with kernel size of 1 that creates a bottleneck by reducing the number of channels, a regular convolutional layer that performs the actual convolution with the compressed feature representation, and another convolutional layer that restores the number of channels. For each residual block, the $ReLU$ activation of the third convolutional layer is applied after addition with the skip connection. Then, an Efficient Channel Attention (ECA) [43] mechanism is applied to select the most discriminative features. Different from the simple channel attention that calculates the importance of channels with a fully-connection layer which implies that no assumption about the topology and relations among bipolar EEG channels is made *a prior*, ECA module uses a one-dimensional convolution layer which enforces locality amongst the channels of deep feature, as the channels of feature at that depth is learnt and do not correspond to EEG channels. After a drop-out layer with rate 50%, a max pooling layer and other two convolutional layers, a global pooling layer is attached to collapse channels of features into an embedding vector.

TABLE I
HYPERPARAMETERS OF LAYERS IN THE EMBEDDING MODULE

No.	layer	units	size	stride	act	others
	Single-Channel Embedding Module					
1	convolutional layer	4	25	5	$ReLU$	L2, BN
2	max-pooling	-	2	2	-	-
3	convolutional layer	8	12	3	$ReLU$	L2, BN
4	convolutional layer	15	15	3	$ReLU$	L2, BN
5	max-pooling	-	2	2	-	-
6	drop-out	-	-	-	-	-
7	linear layer	40	-	-	-	-
	Multichannel Embedding Module					
1	channel-wise conv.	x4	64	8	$ReLU$	L2, BN
2	channel attention	-	-	-	-	-
3	residual block	-	-	-	-	-
3.1	convolution layer	24	1	1	$ReLU$	L2, BN
3.2	convolution layer	24	7	1	$ReLU$	L2, BN
3.3	convolution layer	72	1	1	-	L2, BN
3.4	addition	-	-	-	$ReLU$	-
4	residual block	-	-	-	-	-
5	residual block	-	-	-	-	-
6	ECA Attention	-	-	-	-	-
7	drop-out	-	-	-	-	-
8	max-pooling	-	4	4	-	-
9	convolution layer	128	8	1	$ReLU$	L2, BN
10	convolution layer	256	8	1	$ReLU$	L2, BN
11	global pooling	-	-	-	-	-

Hyperparameters including numbers and sizes of kernels of convolutional layers and max-pooling layers are listed in Table I for both embedding modules. "Units" denotes the number of convolution kernels for convolutional layers and the number of neurons for the linear layer. "Size" denotes kernel size for convolutional layers and pooling size for max-pooling layers. "Stride" refers to the stride of the sliding window for convolutional layers and max-pooling layers. Readers unfamiliar with convolutional neural networks may refer to [16], [17] for a better explanation of the principle and functionality of convolutional layers and pooling. "Act" and "Others" stand for activation function and other hyperparameters, including "L2" indicates the use of $l2$ kernel regularizer and "BN" indicates the use of batch normalization. The drop-out rate of the drop-out layer is set to 40%. A kernel size that equals to half the sampling rate was recommended by [44]. In contrast, smaller kernels were adopted in [3], [38]. We adopted the convention of increasing kernel numbers for deeper layers which is common for CNNs, and larger kernel sizes as [44] suggested.

C. The Metric Module

The metric module takes two embedding vectors as inputs and outputs the similarity based on their weighted distance in the embedding space. The L1 element-wise distance $\boldsymbol{d}$ is first calculated as

$$\boldsymbol{d} = \left\| \boldsymbol{y}_{(1)} - \boldsymbol{y}_{(2)} \right\|_1 = abs(\boldsymbol{y}_{(1)} - \boldsymbol{y}_{(2)}) \tag{1}$$

given embedding vectors $\boldsymbol{y}_{(1)}$ and $\boldsymbol{y}_{(2)}$, where $abs(\cdot)$ denotes element-wise absolute value function given arbitrary vector. A single perceptron with learnable weights $\boldsymbol{w}$ and bias b is used to get a weighted distance d which is a scalar

$$d = \boldsymbol{w}^{\intercal}\boldsymbol{d} + b. \tag{2}$$

Finally, a $sigmoid$ activation is applied converting the distance to a probability between 0 and 1 as the estimation of similarity score.

D. Training Strategy

Unlike an ordinary classification model, the metric learning model cannot directly output the predicted label of category. Instead, the similarity between two input samples is given. A stage-wise training strategy [33] is employed to obtain an end-to-end classification model.

In the stage-wise training strategy proposed in [33], the feature extraction layers were trained by optimizing the triplet loss [28] in the first stage. Similarly in our work, the embedding module is trained by optimizing the cross-entropy loss of similarity, as shown in Algorithm 1. Random pairs of samples are iteratively picked from the training set and used to train the model in this stage.

In the second stage, One of the embedding modules is extracted and locked, and additional three fully-connected layers called classification layers are attached afterward and optimized by minimizing cross-entropy loss of classification, whereas the other embedding module is discarded along with the metric module. Hyperparameters of the classification layers are listed in Table II.

In the third stage, the embedding module is unlocked and the entire model is fine-tuned as a whole, after which an accurate end-to-end real-time seizure detection can be performed with the trained model.

E. Majority Vote Strategy

For the Bonn dataset in which each original recording is divided into 5 segments, a majority vote strategy is employed to obtain the final decision on the original segment, similar to the vote strategy used in [38]. For the CHB-MIT dataset, majority vote is conducted by averaging decisions from 7 overlapped segments.

Algorithm 1: Three-staged training: Stage I

Data: training segments of EEG signals X_{train} and ground-truth labels T_{train}; max iterations I_{max}
Result: trained parameters of embedding module θ_e
Stage I: Metric Learning
construction of embedding module $Embed$ and random initialization of its parameters θ_e;
construction of metric module $Metric$ and random initialization of its parameters θ_m;
for $i \leftarrow 0$ **to** I_{max} **do**
 /* draw random pair of samples $x_{(1)}$, $x_{(2)}$ from X_{train} and their ground-truth labels $t_{(1)}$, $t_{(2)}$ from T_{train} */
 $x_{(1)} \leftarrow random\_pick(X_{train})$;
 $t_{(1)} \leftarrow get\_label(T_{train}, x_{(1)})$;
 $x_{(2)} \leftarrow random\_pick(X_{train})$;
 $t_{(2)} \leftarrow get\_label(T_{train}, x_{(2)})$;
 /* get ground-truth label of similarity given labels of sample pair */
 if $t_{(1)} = t_{(2)}$ **then**
  $t \leftarrow 0$;
 else
  $t \leftarrow 1$;
 end
 /* get embedding vectors of inputs by passing them through embedding module */
 $y_{(1)} \leftarrow Embed(x_{(1)})$;
 $y_{(2)} \leftarrow Embed(x_{(2)})$;
 /* get predicted similarity through metric module */
 $y \leftarrow Metric(y_{(1)}, y_{(2)})$;
 /* calculate the binary cross-entropy (BCE) loss */
 $\mathcal{L} \leftarrow BCE(t, y) \triangleq -[t \log y + (1-t)\log(1-y)]$;
 /* optimize θ_e and θ_m using Adam optimizer */
 $\nabla\theta_e \leftarrow calculate\_gradient(\mathcal{L}, \theta_e)$;
 $\theta_e \leftarrow adam\_optimize(\theta_e, \nabla\theta_e)$;
 $\nabla\theta_m \leftarrow calculate\_gradient(\mathcal{L}, \theta_m)$;
 $\theta_m \leftarrow adam\_optimize(\theta_m, \nabla\theta_m)$;
end

III. Experiments and Results

A. Performance Measures and Evaluation Procedure

The performance of our method was evaluated using well-known and broadly used performance metrics for binary classification including accuracy, specificity, and sensitivity, defined as

$$accuracy = \frac{TP+TN}{P+N} = \frac{TP+TN}{TP+FP+TN+FN} \tag{3}$$

$$sensitivity = \frac{TP}{P} = \frac{TP}{TP+FN} \tag{4}$$

and

$$specificity = \frac{TN}{N} = \frac{TN}{TN+FP}. \tag{5}$$

where P and N denotes positive i.e. ictal and negative i.e. interictal samples, and TP (true positive), FP (false positive), TN (true negative), FN (false negative) means "being ictal and correctly detected to be ictal," "being interictal but wrongly identified as being ictal," "being interictal and correctly detected as interictal" and "being ictal but wrongly classified as interictal," respectively.

On the Bonn dataset, 10-fold cross-validation was used in all experiments mentioned in the paper. Samples are randomly

TABLE II
HYPERPARAMETERS OF CLASSIFICATION LAYERS

layer	units	activation	regularizer
1	64	ReLU	L2
2	8	ReLU	L2
3	1	$sigmoid$	L2

divided into ten equal portions. Nine out of the ten portions are combined and used as the training set whereas the remaining being test set. Then, preprocessing is carried out and the experiment is conducted. This is repeated ten times by choosing every training-versus-test combination. Additionally, the experiment is repeated 5 times for each fold with averaged measures recorded. For stochastic methods like deep learning, the use of repeated experiments is also important on such small dataset in order to obtain a reasonable result on small datasets, otherwise optimal results tend to be inadvertently chosen from multiple attempts as the tuning of some hyperparameters may be mistakenly believed to be the reason of the improvement. Unless otherwise mentioned, experiments are conducted on subsets D versus E classification problem, as they are ictal and interictal intracranial EEG from the same brain location and therefore less influenced by other factors.

On the CHB-MIT dataset, subject-independent evaluation is performed via leave-one-out cross-validation. For each fold, data from one subject is used as test set, data from two other subjects are used as validation set for early stopping of training in case the accuracy on the validation set does not improve for 5 epochs, and data of the remaining subjects are used as training set. Though the model is not explicitly trained on the validation set, decisions are made according to the performance on the validation set, so that only the performance on the test set is reported. Data of every subject is used as test set once and only once.

B. Comparison With Baseline Model

Comparison with baseline models was made to validate the relative improvement of our deep metric learning method, as shown in Table III and Table IV. To ensure the comparison unbiased as much as possible, two baseline models were chosen to be identical to the embedding module used in our deep metric learning models for single-channel and multichannel cases, except for the last several layers. $ReLU$ activation and bias were used in the linear layer, contrary to that of the embedding module. Following that, an output layer is attached producing $sigmoid$-activated estimation instead of embedding vectors. Besides, SVM with rbf kernel is chosen as another baseline model for multichannel signal for reference.

The single-channel baseline model was trained with Adam optimizer of learning rate $1e-3$ for 10 epochs and then with learning rate $1e-4$ for 190 epochs. The deep multichannel baseline model was trained with Adam optimizer of learning rate $1e-3$ for 50 epochs.

Comparably, the models of the proposed method was trained via the stage-wise strategy. For the proposed method on Bonn dataset, both the results without and with majority vote are presented. Compared to the baseline model, our proposed model reached an average accuracy of 97.9% and an average specificity of 99.2% which are 4% and 3.5% higher on the Bonn dataset. In addition, by using majority vote, the average accuracy is improved to 98.60% and the specificity is stabilized at 100%. On the CHB-MIT dataset, the accuracy and sensitivity are 4.5% and 9% higher than the deep learning baseline, and much higher than the SVM baseline.

C. Results on Different Subsets

Another experiment classifying subset C versus E of the Bonn dataset was conducted to verify the generalization ability of our proposed method. The mean and standard deviation of accuracy, sensitivity, and specificity are 0.9848±0.0166, 0.974±0.0338, and 0.9956±0.0068. which are slightly higher, as signals from subset C were recorded from a location different than those from subset D and E and are easier to be discriminated against.

D. Results With Different Metrics

Besides the L1 metric mentioned in Section II, several other metrics and pseudo-metrics were also tested as alternatives in the metric module on the Bonn dataset. They are squared L2 metric which is the square of ordinarily used L2 metric, cosine pseudo-metric which is suitable for higher-dimensional spaces such as the space of word embeddings, and a multilayer perceptron (MLP) metric consists of two fully-connected layers, each defined as

$$d_{l2} = \sum_{i=0}^{\dim(y_{(1)})} w_i^{\intercal}\left(y_{(1)i} - y_{(2)i}\right)^2 + b \tag{6}$$

$$\boldsymbol{d}_{cos} = 1 - \frac{\boldsymbol{y}_{(1)} \odot \boldsymbol{y}_{(2)}}{\left|\boldsymbol{y}_{(1)}\right| \left|\boldsymbol{y}_{(2)}\right|} = 1 - \frac{\boldsymbol{y}_{(1)} \odot \boldsymbol{y}_{(2)}}{1 \times 1} = 1 - \boldsymbol{y}_{(1)} \odot \boldsymbol{y}_{(2)} \tag{7}$$

$$d_{cos} = \boldsymbol{w}^{\intercal}\boldsymbol{d}_{cos} + b \tag{8}$$

and

$$d_{MLP} = MLP(Concatenate(\boldsymbol{y}_{(1)}, \boldsymbol{y}_{(2)})) \tag{9}$$

where $dim(\cdot)$ denotes dimension given vector. Majority vote often causes the scores on some measures to reach 100% and the different conditions cannot be well compared when the scores are all close to 100%. Therefore, majority vote was not applied in this experiment and experiments below. As shown in the upper part of Table V, slightly better performance is achieved with the L1 metric, but the difference between the highest (0.9788 with L1) and lowest averaged accuracy (0.9744 with L2) is less significant than the standard deviation of any experiment. The results further validated the flexibility of the proposed model.

E. Alternative Training Strategy With Classical Classifiers

Other than the three-staged strategy, the trained embedding module can also be used as a deep feature extractor for classical classifiers. Experiments were conducted on the Bonn dataset with SVM, k-NN classifier, decision tree, and Naive Bayes (NB) classifier separately. The SVM was equipped with a linear kernel. The number of neighbors was chosen to be 10 for the k-NN

TABLE III
COMPARING BASELINE MODEL AND THE PROPOSED METHOD ON THE BONN DATASET

model	measure	i-th fold of ten-fold cross-validation										mean ± std
		1	2	3	4	5	6	7	8	9	10	
baseline	accuracy	0.936	0.97	0.942	0.958	0.94	0.964	0.876	0.952	0.97	0.89	0.9398 ± 0.0308
	sensitivity	0.904	0.972	0.944	0.96	0.888	0.952	0.764	0.936	0.98	0.936	0.9236 ± 0.0596
	specificity	0.968	0.968	0.94	0.956	0.992	0.976	0.988	0.968	0.96	0.844	0.956 ± 0.04
proposed method (without majority vote)	accuracy	0.986	0.994	0.954	0.978	0.96	0.988	0.984	0.996	0.948	1	0.9788 ± 0.0175
	sensitivity	0.98	1	0.916	0.956	0.956	0.976	0.968	0.992	0.916	1	0.966 ± 0.0291
	specificity	0.992	0.988	0.992	1	0.964	1	1	1	0.98	1	0.9916 ± 0.0112
proposed method (with majority vote)	accuracy	1	1	0.96	0.98	0.97	1	1	1	0.95	1	0.986 ± 0.0185
	sensitivity	1	1	0.92	0.96	0.94	1	1	1	0.9	1	0.972 ± 0.0371
	specificity	1	1	1	1	1	1	1	1	1	1	1 ± 0

TABLE IV
COMPARING BASELINE MODELS AND THE PROPOSED METHOD ON THE CHB-MIT DATASET

model	measure	i-th fold of leave-one-out cross-validation												
		1	2	3	4	5	6	7	8	9	10	11	12	13
baseline: SVM	accuracy	0.92	0.91	0.89	0.66	0.93	0.395	0.89	0.86	0.91	0.66	0.91	0.48	0.49
	sensitivity	0.882	0.88	0.932	0.396	1	0.526	0.935	0.87	1	1	0.918	0.059	0.255
	specificity	0.959	0.94	0.857	0.904	0.848	0.263	0.852	0.852	0.83	0.346	0.902	0.918	0.778
baseline: deep learning	accuracy	0.941	0.89	0.943	0.865	0.782	0.724	0.926	0.699	0.938	0.938	0.919	0.625	0.5
	sensitivity	0.969	0.78	0.922	0.829	0.994	0.5	0.884	0.542	0.988	0.946	0.854	0.312	0.152
	specificity	0.913	1	0.965	0.901	0.571	0.947	0.968	0.856	0.887	0.93	0.983	0.937	0.848
proposed method	accuracy	0.957	0.890	0.931	0.928	0.911	0.776	0.952	0.757	0.918	0.888	0.968	0.726	0.660
	sensitivity	0.973	0.811	0.922	0.923	0.970	0.692	0.956	0.543	0.967	0.975	0.970	0.481	0.368
	specificity	0.942	0.968	0.940	0.934	0.852	0.860	0.949	0.972	0.869	0.800	0.967	0.970	0.953

model	measure	i-th fold of leave-one-out cross-validation										mean ± std
		14	15	16	17	18	19	20	21	22	23	
baseline: SVM	accuracy	0.477	0.47	0.528	0.55	0.65	0.91	0.67	0.61	0.91	0.88	0.7200 ± 0.1904
	sensitivity	0.227	0.019	0.056	0.133	0.354	0.816	0.34	0.24	0.939	0.804	0.5905 ± 0.3657
	specificity	0.727	1	1	0.891	0.923	1	0.962	0.98	0.882	0.959	0.8510 ± 0.1866
baseline: deep learning	accuracy	0.239	0.502	0.778	0.866	0.863	0.913	0.777	0.605	0.949	0.773	0.7807 ± 0.1829
	sensitivity	0.341	0.005	0.667	0.756	0.758	0.841	0.627	0.211	1	0.909	0.6864 ± 0.2986
	specificity	0.136	0.998	0.889	0.977	0.967	0.986	0.928	1	0.898	0.636	0.8748 ± 0.1931
proposed method	accuracy	0.668	0.746	0.870	0.917	0.967	0.940	0.914	0.825	0.923	0.902	0.8668 ± 0.0959
	sensitivity	0.488	0.523	0.771	0.954	0.935	0.880	0.865	0.676	0.846	0.829	0.7964 ± 0.1921
	specificity	0.849	0.968	0.969	0.880	1	1	0.964	0.974	1	0.975	0.9371 ± 0.0562

TABLE V
COMPARISON WITH DIFFERENT METRIC FUNCTIONS AND WITH DIFFERENT TRADITIONAL CLASSIFIERS

metric/classifier		accuracy	sensitivity	specificity
metric	L1	0.9788	0.9688	0.994
	L2	0.9744	0.9584	0.9904
	cosine	0.9768	0.9584	0.9952
	net	0.9772	0.9584	0.996
classifier	SVM	0.9740	0.9544	0.9936
	k-NN	0.9742	0.9556	0.9928
	decision tree	0.9722	0.9516	0.9928
	Naive Bayes	0.9704	0.9524	0.9884

classifier and the L1 metric is adopted corresponding to that used during metric learning. Gini impurity was used as the criterion of the decision tree and Gaussian prior was selected for the NB classifier. The results are shown in the lower part of Table V. No significant improvements can be observed, indicating that the performance of the proposed method is mainly up to the deep metric learning stage, and the classification layers were not over-fitted more than classical classifiers.

IV. DISCUSSION

According to the results, the proposed method has achieved novel performance for the ictal versus interictal problem on the Bonn dataset, including a 100% specificity on average with majority vote which guarantees no false alarm and high sensitivity of 97.20% on average indicates that most of the seizure activities can be accurately detected. In reality, patients would not be experiencing seizure activities during most of their time and even a specificity of 99.9% translates to 3.7 false alarms per day on average which can be disturbing. An accuracy of 100% can often be reached, though the averaged accuracy of repeated runs is usually lower, meaning the stability of the method still needs some improvements before being capable for real-world clinical applications. Fig. 3 shows the changing of accuracy and loss throughout the training of two runs. Training of the former run shown in Fig. 3(a) and 3(b) converged perfectly at test accuracy

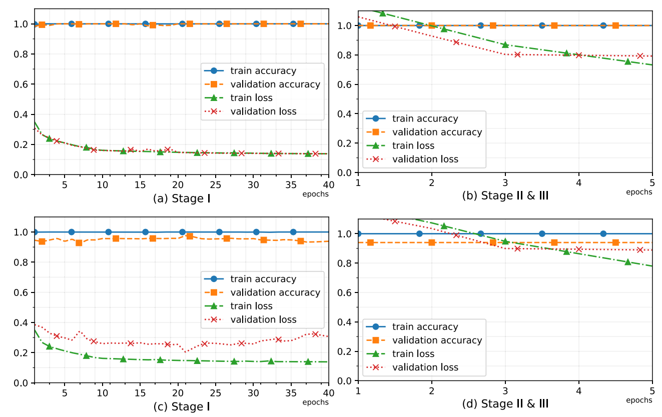

Fig. 3. Accuracy and loss changes over epochs on the Bonn dataset. (a) and (b) shows the result of a good case where the accuracy is 100%. (c) and (d) show the results of a bad case where the model was over-fitted on training set.

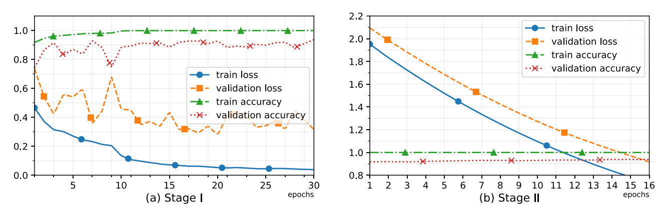

Fig. 4. Accuracy and loss changes over epochs on the CHB-MIT dataset.

100% whereas another run shown in Fig. 3(c) and 3(d) did not. It is clear that the number of training epochs in the first stage is much redundant as the accuracy seldom changes after one or two epochs and the loss also has no significant change. Statistically, however, the redundant epochs give a slightly better result because sometimes the model converges slower and sometimes the accuracy curve tends to be unstable during the first several epochs.

On the CHB-MIT dataset, the accuracy is lower, because scalp EEG has much lower SNR, and neural patterns of seizures can vary significantly across subjects, so that the generalizabiliy is weaken, as shown in Fig. 4.

It can also be pointed out that once the model is over-fitted and the test accuracy never managed to get to 100% during the first stage, the final performance is then limited, irrelevant to the choice of classification layers or classifier in the remaining stages. Otherwise, the embedding vectors can be easily discriminated by classification layers or classical classifiers. This is confirmed through the visualization of embedding vectors. Fig. 5 shows all the embedding vectors outputted by the trained embedding module from the whole subsets D and E of Bonn dataset. Principal component analysis (PCA) is utilized to visualize the 40-dimensional embedding vectors with the most discriminative two components kept. After the embedding module being trained, samples are projected into two separated small regions. As shown in Fig. 5(a) and 5(b), the well-trained embedding module produces embedding vectors that can be easily separated whereas the imperfect module misclassified several ictal segments of signals as interictal. This is due to that just like most of the deep learning models, the proposed model is

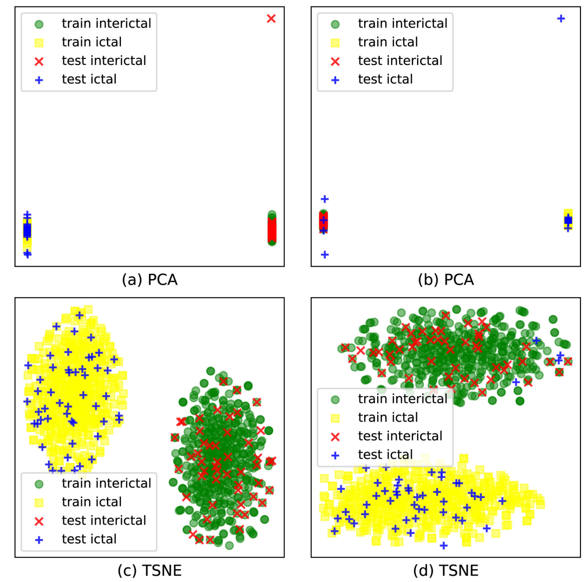

Fig. 5. Visualization of embedding vectors extracted by the embedding module trained in two runs on the Bonn dataset, one good and one imperfect. (a) and (b) are visualization of embedding vectors extracted in the previously mentioned two runs via PCA. (c) and (d) are t-SNE visualization in order to get a clearer view.

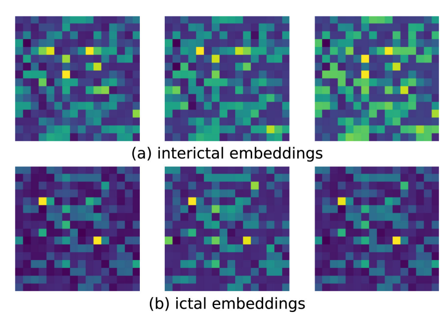

Fig. 6. Visualization of several embedding vectors extracted by the embedding module on CHB-MIT dataset for example. (a) from signals of interictal class. (b) from signals of ictal class.

over-confident about the estimation because it is not designed and trained to be honest about uncertainty. As a result, the additional classifier has no way to correct the errors as the wrongly embedded samples are already surrounded by samples of another class. The problem worth being investigated in future researches to produce a more accurate and more interpretable result. To have a clearer view without the points being overlapped, Fig. 5(c) and 5(d) demonstrated the embedding vectors transformed via t-distributed stochastic neighbor embedding (t-SNE), which is a non-linear visualization and manifold learning method. Fig. 6 shows several embedding vectors extracted from the CHB-MIT dataset for example. To better comfort to human eyes, the embedding vector of 256 dimension is resized to 16 by 16 matrix. The different between embedding vectors of different classes is significant, whereas intra-class variety is largely eliminated. This way, classification can be easily performed, and an obvious drawback is that the generalizability of the model is limited by the lack of diversity of embedding vectors. Future works could try to mitigate the problem by introducing regularizations on the distribution of

TABLE VI
PERFORMANCE COMPARISON AMONG EXISTING METHODS AND THE PROPOSED METHOD ON THE BONN DATASET

Authors	Year	Category	Method	Result
Sharmila et al., [45] *	2016	ML	discrete wavelet transform (DWT) + naive Bayes + k-NN classifier	accuracy 95.62% sensitivity 94.4% specificity 98.51%
Song et al., [46] *	2016	ML	Mahalanobis-similarity-based feature extraction method + extreme learning machine (ELM)	accuracy 97.53% sensitivity 96.18% specificity 98.89%
Bhattacharyya et al., [47] *	2017	ML	Tunable-Q Wavelet Transform Based Multiscale Entropy Measure	accuracy 98%
Sharma et al., [48] *	2017	ML	analytic time-frequency flexible wavelet transform and fractal dimension	accuracy 98.50% sensitivity 100% specificity 97%
Ahmedt et al., [49] *	2018	DL	LSTM	accuracy 96.50%
Ullah et al., [38] *	2018	DL	pyramidal 1D-CNN + majority vote	accuracy 99.3%
Qiu et al., [50]	2018	DL	Denoising Sparse Autoencoder + logistic regression classifier	accuracy 100% sensitivity 100% specificity 100%
Thara el at., [19]	2019	DL	stacked bidirectional LSTM	accuracy 93% precision 95% recall 96%
Ilakiyaselvan et al., [51] *	2020	DL	Deep learning with reconstructed stage space images	accuracy 97.78%
this work *	-	DL	proposed method (without majority vote)	accuracy 97.88% sensitivity 96.60% specificity 99.16%
			proposed method (with majority vote)	accuracy 98.60% sensitivity 97.20% specificity 100%

TABLE VII
PERFORMANCE COMPARISON AMONG EXISTING METHODS AND THE PROPOSED METHOD ON THE CHB-MIT DATASET

Authors	Year	Cross-Validation	Method	Result
Yao et al., [52]	2021	leave-one-out cross-validation	spatial attention + BiLSTM	accuracy 83.89% sensitivity 83.72% specificity 84.06%
Wei et al., [53]	2019	leave-one-out cross-validation	Wasserstein GAN + MIDS	accuracy 84.00% sensitivity 72.11% specificity 95.89%
Jiang et al., [54]	2019	not mentioned	dual-tree discrete wavelet transform + TCA	accuracy 74.03% sensitivity 70.17% specificity 77.90%
Ansari et al., [55]	2020	leave-one-out cross-validation	band power + neutrosophic logic based k-NN	sensitivity 85% specificity 89.06%
this work	-	leave-one-out cross-validation	proposed method	accuracy 86.68% sensitivity 79.64% specificity 93.71%

embedding vectors or incorporating adversial learning, domain adaption, or other related methods and strategies.

There are many existing methods that aim at automatic seizure detection using machine learning and their performance is also impressive. Comparison of performance among several existing methods on the same datasets and our proposed method is shown in Table VI and Table VII. "Category" denotes the category of method whereas "ML" and "DL" refer to works that only used traditional machine learning methods and works that involves deep learning methods, respectively. Performance of existing methods are those reported in their corresponding papers. Only those results from experiments that are explicitly stated as or can be inferred to be binary classification are included for comparison on the Bonn dataset, among which results known to be of D vs E classification are marked with superscript "*". According to [56] it is most difficult for seizures to be detected in these two subsets. Only results of subject-independent works are included for comparison on the CHB-MIT dataset.

On the Bonn dataset, Sharmila *et al.*, [45] used discrete wavelet transform (DWT) to capture the feature of signal at multiple scales, and naive Bayes and k-NN for classification. Song *et al.*, [46] also utilized DWT, along with Mahalanobis-similarity-based and sample-entropy-based feature extraction methods. Our method has some commonalities with the Mahalanobis-similarity-based method. But differently, the metric function of our model can be view as generalized Mahalanobis distance in an learnt embedding space. The embedding module is also a learnt feature extractor compared to DWT and other handcrafted feature extraction methods and achieved higher accuracy. Bhattacharyya *et al.*, [47] extracted k-NN entropy feature based on wavelet transform with tunable quality factor and the accuracy was 98%. Sharma *et al.*, [48] achieved 100% sensitivity by using analytic time-frequency flexible wavelet transform (ATFFWT) which is an improved wavelet transform, and fractal dimension feature that can characterize non-stationary signals well. Ahmedt *et al.*, [49] proposed an early work that employed a LSTM which is a basic deep learning model for seizure detection based on raw signal and got accuracy of 96.5% which is comparable to traditional machine learning methods. Ullah *et al.*, [38] borrowed the idea of pyramidal CNN architecture from Computer Vision and proposed an one-dimensional version for raw EEG signal, which further validated the effectiveness of deep models on small datasets. Qiu *et al.*, [50] used a combination of denoising autoencoder and sparse autoencoder which are both successful self-supervised deep learning models as learnable feature extractor. Together with a simple logistic regression classifier, their model achieved 100% accuracy, showing that the combination of traditional machine learning models and deep models may improve the performance by utilizing the advantages of both. Self-supervision is also one of the ways to overcome the lack of annotated data, and autoencoders can be naturally integrated along with other methods including deep metric learning and domain adaptation which may be a promising direction, as they both involves the use of embedding or latent space and the separation of deep feature extraction and classification. Besides, Thara *et al.*, [19] utilized stacked bidirectional LSTM, and Ilakiyaselvan *et al.*, [51] constructed stage space images from EEG signal and used common backbone networks including GoogLeNet. Among those works, many mentioned the use of k-fold cross-validation whereas few reported the use of repeated experiments, without which 100% can often be reached. Additionally, other factors like the length of segments also vary. Therefore, the comparison is for reference only and it is recommended to focus more on methods themselves.

On the CHB-MIT dataset, Yao *et al.*, [52] leveraged spatial attention and bidirectional long short-term memory (BiLSTM), resulted in an impressive average accuracy of 83.89% for cross-patient validation. Wei *et al.*, [53] achieved accuracy of 84% by using Wasserstein GAN for data augmentation, and merger of the increasing decreasing sequences (MIDS) to highlight the features. Jiang *et al.*, [54] used dual-tree discrete wavelet transform for feature extraction and performed Transfer Component Analysis (TCA) to reduce individual difference as an attempt of transferable learning for traditional feature extraction. Ansari *et al.*, [55] extracted band power feature and used neutrosophic logic based k-NN for classification. They also used genetic algorithm for searching of optimal parameters and further improved their specificity to 89.06%.

V. Conclusion

In this study, an automatic seizure detection method has been proposed and tested, including embedding modules, metric module, and corresponding stage-wise training strategy of the model. An optional majority vote strategy is also employed for experiments on the Bonn dataset. With the proposed method, accurate seizure detection is done by tackling the lack of data through deep metric learning. The trained model is small and can be easily deployed to wearable devices with limited resources for real-world scenarios. Also, the proposed method is general and can be easily employed for more tasks involving EEG signals and other biomedical recordings such as magnetoencephalogram (MEG), electrocardiogram (ECG), and electromyogram (EMG) with minor tweaks of the hyper-parameters. No extra feature engineering is required.

There are some limitations to our model including the need of class balance, over-confidence on wrong classifications, and the necessary improvements on interpretability. By undersampling the interictal-ictal ratio to 1:1 by randomly discard interictal samples, the method currently does not take full advantage of the training data. Future works could consider adopting methods such as triplet focal loss [57] into deep metric learning framework to better utilize the available data. Moreover, various other improvements can be made. For instance, the method needs to be improved and tested on more datasets and tested for cross-dataset validation which is important for practice use. Secondly, methods and techniques like Bayesian approaches can be applied to address the problem of over-confidence mentioned in Section IV. Besides, both deep learning and deep metric learning are still emerging and promising collections of methods and the possible improvements are abundant, including the incorporation of other closely relevant methods like domain adaption and generalization, self- and semi-supervised learning using autoencoders, and adversial learning to further tackle the impact of inter-subject variance.

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